

Causal Machine Learning: Homework 1

Due date: September 22, 11:59PM

Your name:

Bias-Variance Tradeoff

1. In your own words, explain the bias–variance tradeoff when using a *sample* to approximate the *population* Conditional Expectation Function (CEF), and clarify where the variance in prediction comes from.
2. In your own words, explain how *Random Forests* address the bias–variance tradeoff, and why balancing this tradeoff is important.
3. In your own words, explain how *Boosted Trees* differ from *Bagging* and *Random Forests* in how they address the bias–variance tradeoff.

Frisch–Waugh–Lovell Theorem

In lecture, we showed that the multivariate OLS regression of Y_i against the treatment variable D_i and control covariates W_i

$$Y_i = \beta_1 D_i + \beta_2' W_i + \epsilon_i$$

can be recovered in a bivariate regression,

$$\begin{aligned}\tilde{Y}_i &= \beta_1 \tilde{D}_i + \mu_i \\ \beta_1 &= \frac{\text{Cov}(\tilde{Y}_i, \tilde{D}_i)}{\text{Var}(\tilde{D}_i)}\end{aligned}$$

where $\tilde{D}_i$ and $\tilde{Y}_i$ are defined as residuals from their respective regressions against W_i .

$$\begin{aligned}\tilde{D}_i &= D_i - \gamma'_{DW} W_i, & \tilde{Y}_i &= Y_i - \gamma'_{YW} W_i \\ \gamma_{DW} &= \arg \min_{\gamma} E \left[(D_i - \gamma' W_i)^2 \right], & \gamma_{YW} &= \arg \min_{\gamma} E \left[(Y_i - \gamma' W_i)^2 \right]\end{aligned}$$

1. Show that β_1 can be similarly recovered by using Y_i rather than residualized $\tilde{Y}_i$

$$\beta_1 = \frac{\text{Cov}(Y_i, \tilde{D}_i)}{\text{Var}(\tilde{D}_i)}$$

2. Symmetrically, can one use unresidualized D_i in the numerator to express β_1 ?

$$\beta_1 \stackrel{?}{=} \frac{\text{Cov}(\tilde{Y}_i, D_i)}{\text{Var}(\tilde{D}_i)}$$

3. In a high-dimensional setting where $p/n \not\rightarrow 0$, the OLS estimator of β_1 is no longer well-behaved, and its sampling variability can be severely inflated. We introduced Double LASSO in lecture to address this high-dimensional problem.
- (a) Explain in words the relationship between the *penalty level* λ in Double LASSO and the theoretical variance of your estimated β_1 .
- (b) After running Double LASSO, we obtain residualized D_i and Y_i by partialling out high-dimensional controls. In the final bivariate regression of residualized Y_i on residualized D_i , can we rely on the standard sandwich formula for the standard error of $\hat{\beta}_1$? Why or why not?

Test of Convergence Hypothesis

In lecture, we provided an empirical example of partialling-out with Double LASSO to estimate the regression coefficient β_1 in the high-dimensional linear regression model. Specifically, we estimated how the rates at which economies of different countries grow (Y_i) are related to the initial wealth levels in each country (D_i) controlling for country's institutional, educational, and other similar characteristics (W). The relationship is captured by β_1 , the *speed of convergence/divergence*, which measures the speed at which poor countries catch up ($\beta_1 < 0$) or fall behind ($\beta_1 > 0$) rich countries, after controlling for W_i .

Indeed, residualized D_i and Y_i can also be estimated using Machine Learning (ML) methods beyond LASSO, as demonstrated in lecture. In this question, you will estimate residualized D_i and Y_i using the ML methods introduced in class. For this exercise, we will use the dataset `GrowthData`, which is included in the `hdm` package.

```
library(hdm)
## function to get data
getdata <- function(...) {
  e <- new.env()
  name <- data(..., envir = e)[1]
  e[[name]]
```

```

}

## now load your data calling getdata()
growth <- getdata(GrowthData)

## create the outcome variable y, treatment d, and control covariates W
y <- growth$Outcome
W <- growth[-which(colnames(growth) %in% c("Outcome", "intercept", "gdpsh465"))]
D <- growth$gdpsh465

```

1. Using the package `randomForest` and its default settings of `ntree` and `nodesize`, report the estimated β_1 . Use `set.seed(1)`. What can you conclude from your estimation?
2. Now change the setting and let `ntree = 10` and `nodesize = 1`. Use `set.seed(1)`. Re-estimate β_1 and report your findings. In your own words, explain the possible reasons of changes in your findings.

```

set.seed(1)
library(randomForest)

```

3. We may also use a Neural Network model to estimate the residualized D_i and Y_i . Set the random seed with `set.seed(1)`. Configure the hyperparameters as follows:
 - Build a neural network with one hidden layer containing 200 neurons.
 - Train the network using Stochastic Gradient Descent (SGD) with a `learning rate` of 0.05, 200 `epochs`, and a `batch size` of 10.
 - Implement early stopping based on performance on a 20% validation set, with a patience of 10 `epochs`.

Based on your estimation results β_1 under this model, what can you conclude about the relationship between country's initial wealth and growth rate?

```

set.seed(1)
library(keras3)

```